RUOSHI XU

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PROFILE

Ruoshi Xu's research has focused on **medical image analysis** and **multimodal intelligence**, including robust medical image segmentation (CIRSeg, ACME), world-knowledge reasoning in image generation (WorldGenBench), and 3D geometric reasoning (Point-VLM). His current research interests lie in **the methods and applications of visual intelligence and multimodal agents**.

EDUCATION & EXPERIENCE

Southern University of Science and Technology (SUSTech) **Sep. 2022 – Jun. 2026**
B.Eng. in Intelligence Science and Technology (**Turing Class**) **GPA: 3.90/4.00, Rank: 1/22**
Supervised by Prof. Jianguo Zhang

Hengqin Laboratory (Guangdong Provincial Laboratory) **Jul. 2026 – Present**
Research Intern Supervised by Prof. Jianguo Zhang and Prof. Shengda Luo

PUBLICATIONS & RESEARCH

CIRSeg: Coarse-to-Fine Intensity-Robust Liver Segmentation with Source-Free Continual Test-Time Adaptation [Paper]
Ruoshi Xu, Mingqi Gao, Shengda Luo, Jingkun Chen.
MICCAI 2026 CARE Workshop.
• Led the project end-to-end, including the design and implementation of the segmentation framework, all experiments, manuscript writing, and poster preparation.
• Achieved 97.13%/97.93% Dice and 20.18/11.30 mm HD95 on the ID/OOD test sets (official results), respectively; ranked **2nd among 112 teams on the GED4 validation set**.

ACME: Annotation-Compatible Model-Driven Exploration for Single-Source Domain Generalization in Medical Image Segmentation [Project]
Shengda Luo, **Ruoshi Xu***, Jingkun Chen.
Submitted to IEEE Transactions on Medical Imaging (TMI).
• Designed and implemented the complete Probe–Qualify–Learn framework; led all experimental design and evaluation, authored the Method and Experiments sections.
• Outperformed the best compared methods across four benchmarks, improving Dice by 4.44 pp (Fundus), 2.79/0.13 pp (MSCMR, bidirectional), 7.50 pp (Polyp), and 10.27 pp (Prostate).

WorldGenBench: A World-Knowledge-Integrated Benchmark for Reasoning-Driven Text-to-Image Generation [Paper]
Ruoshi Xu, Daoan Zhang*, ···, Shengda Luo.
CVPR2025 GMCV Workshop.
• Contributed to model evaluation using benchmark metrics and research visualizations; led subsequent manuscript revisions, rebuttal preparation, and resubmission efforts.
• Revealed substantial gaps in world-knowledge implicit reasoning among text-to-image models, with GPT-4o achieving the highest Knowledge Checklist Scores of 24.46 (Humanities) and 19.61 (Nature).

*Equal contribution.

Reinforcing 3D Understanding in Point-VLMs via Geometric Reward Credit Assignment [Paper]
Jingkun Chen, **Ruoshi Xu***, Mingqi Gao, Shengda Luo, Jungong Han.
Submitted to IEEE Transactions on Multimedia (TMM).
• Proposed geometric reward credit assignment, designed and implemented the ShapeNetCore-based synthetic data generation pipeline, and conducted all training and experimental evaluations.
• Improved 3D Key Point Accuracy from 0.64 to 0.93, achieving 0.686 in 3D bounding box IoU and 0.852 in reprojection consistency while preserving robust 2D localization performance.

Beyond Cross-Spectral Complementarity: Semantic-Physical Interaction for RGB-IR Exposure Correction [Project]
Ruoshi Xu, Shengda Luo, Jingchao Peng, Jingkun Chen.
Manuscript in revision.
• Proposed SemPhys-ER, a semantic-physical expert routing framework that combines spatial, frequency, and cross-modal RGB-IR evidence with frozen multi-level SigLIP guidance for extreme exposure correction.
• Achieved 27.29 dB average PSNR and 0.8895 average SSIM on HDRT, improving low-/high-exposure PSNR by 2.58/2.44 dB over the previous RGB-IR method; ranked first on seven of nine LLVIP no-reference metrics.

*Equal contribution.